

From Circuits to Graphs, From Graphs to Space

1

UNESP — LaPSEE Graduate Minicourse

Session 1 — 2 h

- I. Three lenses on a power grid
- II. The grid as a graph: **A**, **L**, **B**
- III. Spectral theory of the grid Laplacian
- IV. Centrality and its discontents
 - Lab 1a — centrality + Motter–Lai
- V. Spatial correlation on the grid
- VI. From correlation to regression
 - Lab 1b — Moran, LISA, OLS-trap

8'

15'

18'

12'

15'

20'

22'

10'

Take-away

One algebraic object — the weighted graph Laplacian **B** — governs the DC power flow (Part II), the way disturbances propagate (Part III), and the spatial dependence of bus-indexed data (Parts V–VI). The session is built around *that* observation.

Algebraic / structural.

1. Assemble $\mathbf{A}$, $\mathbf{L}$, $\mathbf{B}$ from a one-line diagram.
2. Solve $P^{\text{inj}} = \mathbf{B} \boldsymbol{\theta}$ with a slack.
3. Read a PTDF column as a current divider.
4. Interpret $\lambda_2(\mathbf{L})$ and the Fiedler vector.
5. Compute effective resistance $Z_{\text{eff}}(i, j)$ from $\mathbf{L}^\dagger$.

Spatial-statistical.

1. Compute Moran's I given $\mathbf{W}$ and $\mathbf{y}$.
2. Diagnose HH/LL/HL/LH clusters with LISA.
3. Pick a $\mathbf{W}$ from *four* principled candidates and defend it.
4. Recognise when OLS is biased on spatial data.
5. Report direct/indirect/total effects from a SAR fit.

Take-away

Every abstract object is shown first on a hand-checkable 4-bus toy, then on IEEE 30 in the lab. *If you cannot do it on the toy, you do not understand it on IEEE 30.*

Notation in one place

Symbol	Meaning	Lives in
$\mathbf{A}$	adjacency matrix (binary)	$\{0, 1\}^{n \times n}$
$\mathbf{D}, \mathbf{L} = \mathbf{D} - \mathbf{A}$	degree, combinatorial Laplacian	$\mathbb{R}^{n \times n}$
$\mathbf{B} = \mathbf{L}_Y$	susceptance-weighted Laplacian	$\mathbb{R}^{n \times n}$
$\lambda_k(\mathbf{L}), \mathbf{v}_k$	eigenvalues and eigenvectors of $\mathbf{L}$	—
$\mathbf{L}^\dagger$	Moore–Penrose pseudoinverse of $\mathbf{L}$	$\mathbb{R}^{n \times n}$
p^{inj}, θ	nodal injection, voltage angle	$\mathbb{R}^n$
$\mathbf{H}$	power transfer distribution factors	$\mathbb{R}^{L \times n}$
$Z_{\text{eff}}(i, j)$	effective electrical (reactance) distance	$\mathbb{R}_{\geq 0}$
$\mathbf{W}, \tilde{\mathbf{W}}$	spatial weights, row-standardised	$\mathbb{R}^{n \times n}$
I	Moran's global autocorrelation	$[-1, 1]$
ρ, λ	SAR / SEM dependence parameters	$(-1, 1)$

Take-away

n = buses, L = branches. IEEE 30: $n = 30$, $L = 41$. *Indexing convention.* Math is 1-indexed; the pandapower/IEEE-30 frames are 0-indexed (slack = bus 0).

I. Three lenses on a power grid

Pick a bus on the IEEE 30. *Why* does that bus matter?

- Because removing it triggers a cascade? ($\Rightarrow$ Part IV)
- Because a shock there reaches everywhere quickly? ($\Rightarrow$ Part III)
- Because its variable (price, demand) covaries with neighbours? ($\Rightarrow$ Part V)
- Because ΔX at that bus changes y at *other* buses? ($\Rightarrow$ Part VI)

Take-away

Four very different operational meanings of “mattering”. Each is a question we will answer with a *different* object built from the same graph.

Three lenses on the same physical grid

Circuit

Nodes = voltages, edges = impedances. KCL/KVL govern flows. *Local, deterministic.*

Graph

$\mathcal{G} = (\mathcal{V}, \mathcal{E})$; matrices $\mathbf{A}$, $\mathbf{L}$, $\mathbf{B}$ encode *structure* (Pagani and Aiello 2013; Dörfler et al. 2018).

Space

Bus-indexed variables exhibit *statistical dependence* via $\mathbf{W}$ (Anselin and Rey 2014; J. LeSage and Pace 2009).

Key idea. The bridge between the three lenses is a **single matrix**: the susceptance Laplacian $\mathbf{B}$. Its non-zero pattern is the graph (lens 2), its action $\mathbf{B}\boldsymbol{\theta} = P^{\text{inj}}$ is the circuit (lens 1), and one of its spectral cousins becomes the spatial weight matrix $\mathbf{W}$ (lens 3).

- **Session 1.** Structure *and* space: $\mathbf{B}$ as a Laplacian, its spectrum, and $\mathbf{W}$ as the spatial weight matrix.
- **Session 2.** Prices and siting: the LMP is the dual of the DC-OPF nodal balance (Schweppe et al. 1988); its spatial structure is regressed with the tools built today.
- **Session 3.** Behaviour: buses become agents; the same $\mathbf{W}$ becomes their communication graph (Albrecht et al. 2024; Fährnich et al. 2024).

Key idea. Three sessions, one mathematical object. By the end of today, you will have seen $\mathbf{B}$ in five different roles — and the spectrum of *some* matrix will appear in every one of them.

II. The grid as a graph

Definition

A power network is modelled as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with $\mathcal{V} = \{1, \dots, n\}$ the buses and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ the branches (lines or transformers).

Three graph models of the same physical grid:

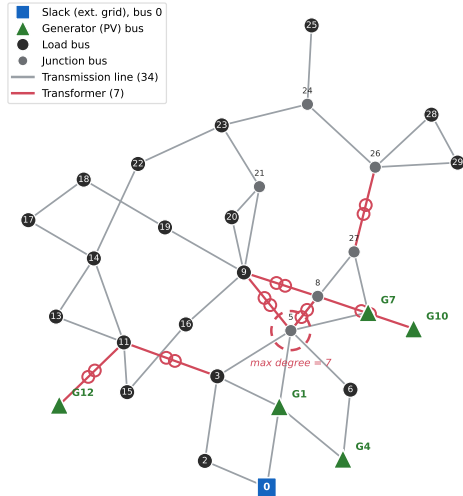
1. **Topological** $\mathcal{G}$: pure connectivity (Pagani and Aiello 2013; Cuadra et al. 2015).
2. **Weighted electrical** $\mathcal{G}_Y$: edges weighted by admittance magnitude $|y_{ij}|$ (Wang et al. 2011).
3. **Directed power-flow** $\mathcal{G}_F$: orientation from $\text{sign}(P_{ij}^{\text{flow}})$ under a reference dispatch (Hines et al. 2010).

Take-away

Choosing a graph model is a research choice. We will use $\mathcal{G}$ (Part II–IV) and a spectral cousin of $\mathcal{G}_Y$ (Parts V–VI).

The benchmark we use all session: IEEE 30

IEEE 30-bus test system — one-line diagram (30 buses, 41 branches; layout from pandapower geodata)



Slack at bus 0; PV generators at buses 1, 4, 7, 10, 12; 21 load buses. 34 lines + 7 transformers = 41 branches (pandapower representation).

$$\mathbf{A}_{ij} = \mathbf{1}[(i, j) \in \mathcal{E}], \quad \mathbf{D} = \text{diag}(d_1, \dots, d_n), \quad \mathbf{L} = \mathbf{D} - \mathbf{A}.$$

- $\mathbf{A} = \mathbf{A}^\top$, sparse, $\{0, 1\}$ -valued. Row sum d_i is the degree.
- $\mathbf{L} = \mathbf{L}^\top \succeq 0$; $\mathbf{L}\mathbf{1} = \mathbf{0}$, so $\lambda_1(\mathbf{L}) = 0$.
- $(\mathbf{A}^k)_{ij}$ counts walks of length k between i and j .

Example (IEEE 30 at a glance)

$n = 30$, $L = 41$, $\bar{d} = 2.73$, $d_{\max} = 7$ at bus 5. Connected (single component); $\lambda_2(\mathbf{L}) = 0.2121$ (computed in Lab 1a; Part III explains what that number means).

Worked example — A, L on a 4-bus toy

Toy grid: triangle {1,2,3} plus a pendant bus 4; edges {12, 13, 23, 34}.

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{D} = \text{diag}(2, 2, 3, 1),$$

$$\mathbf{L} = \mathbf{D} - \mathbf{A} = \begin{bmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

- $\mathbf{L} \mathbf{1} = \mathbf{0}$ ($\mathbf{1}$ is a null vector).
- Spectrum: $\{0, 1, 3, 4\}$ (computed in Part III).
- Cutting edge 34 disconnects bus 4 $\Rightarrow \lambda_2 = 0$.

Take-away

This 4×4 matrix is a complete miniature of the DC power flow apparatus. Anything we do with $\mathbf{B}$ on IEEE 30 we will first justify with a pencil here.

- The bus admittance matrix $\mathbf{Y}_{\text{bus}} \in \mathbb{C}^{n \times n}$ is a weighted adjacency:

$$(\mathbf{Y}_{\text{bus}})_{ij} = -y_{ij}, \quad (\mathbf{Y}_{\text{bus}})_{ii} = \sum_{k \neq i} y_{ik} + y_i^{\text{sh}}.$$

- Off-diagonal non-zero pattern $\Leftrightarrow$ edges of $\mathcal{G}$.
- DC approximation: $\mathbf{B} = -\Im(\mathbf{Y}_{\text{bus}}) = \mathbf{L}_Y$, the **admittance-weighted Laplacian**, with edge weights $b_{ij} = 1/x_{ij}$.

Take-away

$\mathbf{Y}_{\text{bus}}$ is a circuit object and $\mathbf{L}$ is a graph object, but they share a skeleton. $\mathbf{B} = \mathbf{L}_Y$ is the exact bridge (Dörfler et al. 2018).

Dörfler et al. (2018) derive the DC approximation from KCL on the susceptance graph:

$$P^{\text{inj}} = \mathbf{B} \theta$$

with

$$\mathbf{B}_{ii} = \sum_{j \sim i} b_{ij}, \quad \mathbf{B}_{ij} = -b_{ij} \quad ((i, j) \in \mathcal{E}), \quad b_{ij} = \frac{1}{x_{ij}}.$$

Sign convention. $\mathbf{B}$ is a positive-diagonal Laplacian, so $P_i^{\text{inj}} = \sum_{j \sim i} b_{ij}(\theta_i - \theta_j) = (\mathbf{B}\theta)_i$. Equivalently $\mathbf{B} = -\Im(\mathbf{Y}_{\text{bus}})$; the minus sign appears only if you write $\mathbf{B} = \Im(\mathbf{Y}_{\text{bus}})$ with its negative diagonal.

$\mathbf{B}$ has $\mathbf{1}$ in its null space; the system is rank-deficient. Fix the angle reference at the slack bus s : remove its row and column.

$$\theta_{-s} = \mathbf{B}_{-s,-s}^{-1} P_{-s}^{\text{inj}}, \quad \theta_s = 0.$$

Example (IEEE 30, slack at bus 0)

Solving manually and comparing to pandapower (Thurner et al. 2018):

$$\max_i |\theta_i^{\text{manual}} - \theta_i^{\text{pp}}| = 0.298^\circ \quad (\text{at bus 11}).$$

Residual is due to neglected line resistances. Notebook cell 1a.3.

Take-away

The DC power flow is a *Laplacian linear system*. Every property of $\mathbf{B}$ we derive next applies to it.

Worked example — solving $P^{\text{inj}} = \mathbf{B}\theta$ by hand

Same 4-bus toy, unit susceptance $b_{ij} = 1$. Transaction: inject $+1$ at bus 2, withdraw -1 at bus 4; slack = bus 1.

$$\mathbf{B} = \begin{bmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}, \quad P^{\text{inj}} = \begin{bmatrix} 0 \\ +1 \\ 0 \\ -1 \end{bmatrix}.$$

Drop row/col 1, solve $\theta_{-1} = \mathbf{B}_{-1,-1}^{-1} P_{-1}^{\text{inj}}$:

$$\theta = (0, +19.10^\circ, -19.10^\circ, -76.39^\circ).$$

Take-away

Branch flow $f_{ij} = b_{ij}(\theta_i - \theta_j)$. The angle magnitudes are large here because $x = 1$ pu is artificially big — the *flow split* we compute later is scale-free.

Key idea. The matrix that governs the DC power flow *is* the weighted Laplacian of the admittance graph.

- Topology of $\mathcal{G} \rightarrow$ sparsity of $\mathbf{B}$.
- Edge weights $b_{ij} \rightarrow$ magnitudes in $\mathbf{B}$.
- Slack bus $\rightarrow$ reference for θ ; null-space pinning.

Centrality (Part IV), spatial weights $\mathbf{W}$ (Part V), and the LMP of Session 2 are all built on top of $\mathbf{B}$ and its spectrum.

III. Spectral theory of the grid Laplacian

$\mathbf{B}$ is a symmetric positive-semidefinite matrix. By the spectral theorem, every linear question about $\mathbf{B}$ — the DC solve, the propagation of a disturbance, the notion of distance on the grid — has a clean answer in the eigenbasis.

Key idea. Spectrum is not decoration. The three operational meanings of “mattering” from Part I — propagation, robustness, statistical coupling — are each a statement about a particular eigenvalue or eigenvector of $\mathbf{B}$ (or its cousin $\mathbf{W}$).

L symmetric PSD $\Rightarrow$ orthonormal eigenbasis $\{v_1, \dots, v_n\}$ with

$$0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n.$$

- $\lambda_1 = 0$ always, with eigenvector $v_1 \propto \mathbf{1}$.
- $\lambda_2 > 0$ iff $\mathcal{G}$ is connected; this is the **algebraic connectivity** (Fiedler).
- Multiplicity of 0 = number of connected components.
- Largest eigenvalue λ_n bounds spectral radius; for B , governs the worst-case angle response per unit injection.

Take-away

One number (λ_2) tells you whether the grid is connected and *how strongly*. Two numbers (λ_2 and λ_n) bound the conditioning of every DC-OPF solve on this grid.

Worked example — three topologies, three spectra

Four buses, three different edge sets, all with unit weights:

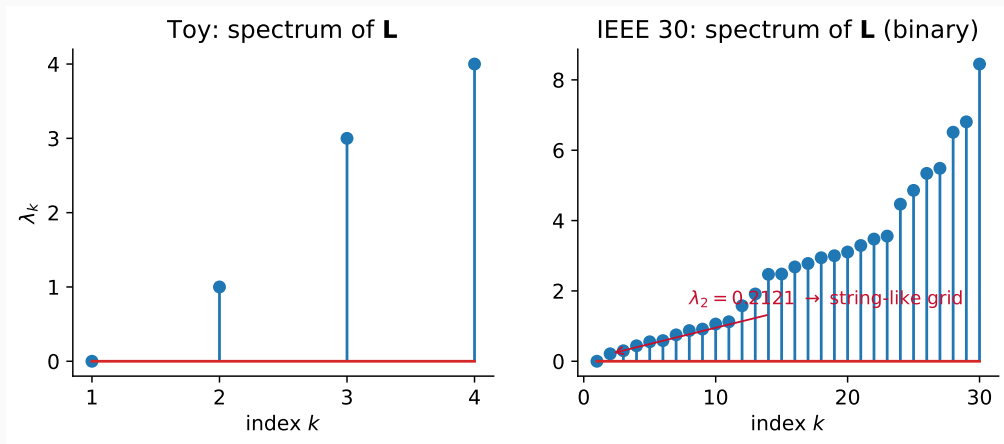
Topology	Edges	Spectrum of $\mathbf{L}$
Path 1–2–3–4	$\{12, 23, 34\}$	$\{0, 0.586, 2.000, 3.414\}$
Ring 1–2–3–4–1	$\{12, 23, 34, 14\}$	$\{0, 2.000, 2.000, 4.000\}$
Star (centre 3)	$\{13, 23, 34\}$	$\{0, 1.000, 1.000, 4.000\}$
Toy of Part II (triangle + pendant)	$\{12, 13, 23, 34\}$	$\{0, 1.000, 3.000, 4.000\}$

- Ring has the *largest* λ_2 (most “connected”).
- Path has the *smallest* non-zero λ_2 for these four nodes — string-like graphs are weakly connected algebraically.
- IEEE 30 is *string-like*: $\lambda_2 = 0.2121$, much closer to a path than to a ring of comparable size.

Take-away

Topology you can draw $\rightarrow$ a number you can compute. λ_2 is the *algebraic* measure of mesh redundancy.

The spectra, visualised: toy vs. IEEE 30



Left: the toy of Part II has 4 eigenvalues $\{0, 1, 3, 4\}$ — $\lambda_2 = 1$ marks a robustly connected mini-graph. Right: IEEE 30 has $\lambda_2 = 0.2121$, an order of magnitude below the toy. The flat cluster of small eigenvalues right after λ_2 is the algebraic fingerprint of a *string-like* grid topology — many directions along which a disturbance is poorly damped.

The Fiedler vector and graph partitioning

Let v_2 be the unit eigenvector for λ_2 (the *Fiedler vector*).

- Sign of $v_2(i)$ defines an (approximate) **bipartition** of $\mathcal{V}$ into two halves with few edges between them.
- Magnitude of $v_2(i)$ measures how “peripheral” bus i is in its half.
- The classical **Cheeger inequality** bounds the bottleneck (edge expansion) of any partition by λ_2 ; a small λ_2 guarantees a near-disconnect.

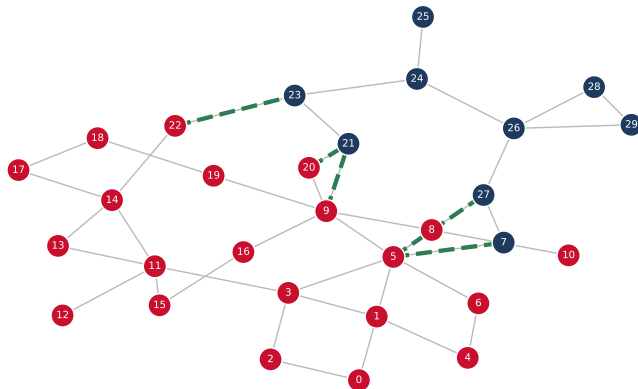
Example (IEEE 30 Fiedler)

Buses with $v_2 > 0$ and $v_2 < 0$ split the system along a natural corridor. Removing the few edges crossing the partition is the cheapest islanding strategy — and a worst-case scenario for a correlated outage. Notebook cell 1a.4.

Take-away

The Fiedler vector is not just a vector — it is the grid's *natural east–west axis*.

IEEE 30 Fiedler bipartition
navy + v_2 (9) | red - v_2 (21) | 5 dashed-green cut branches



Sign of $v_2(i)$ paints buses navy (+) vs. red (-). Dashed green branches cross the partition — only 5 of the 41 branches do. Cutting those 5 isolates the 9-bus minority cluster from the 21-bus majority. The Fiedler vector identifies the network's weakest cross-section algebraically, without any operator input.

Reduced system $\mathbf{B}_{-s,-s} \boldsymbol{\theta}_{-s} = \mathbf{P}_{-s}^{\text{inj}}$ has symmetric PD $\mathbf{B}_{-s,-s} = \sum_{k=1}^{n-1} \mu_k \mathbf{u}_k \mathbf{u}_k^\top$. Project the injection: $\mathbf{P}_{-s}^{\text{inj}} = \sum_k \langle \mathbf{u}_k, \mathbf{P}_{-s}^{\text{inj}} \rangle \mathbf{u}_k$. Then

$$\boldsymbol{\theta}_{-s} = \sum_{k=1}^{n-1} \frac{\langle \mathbf{u}_k, \mathbf{P}_{-s}^{\text{inj}} \rangle}{\mu_k} \mathbf{u}_k.$$

- Each mode $\mathbf{u}_k$ contributes *independently*; the response is amplified by $1/\mu_k$.
- **Modes with small μ_k dominate**: a near-disconnect ($\mu_2 \approx \lambda_2(\mathbf{B}_{-s,-s})$ small) makes the system extremely sensitive to injections that align with $\mathbf{u}_2$.
- This is the spectral statement of the rule of thumb “stress concentrates at bottlenecks”.

Take-away

The DC power flow is, eigenvector by eigenvector, a decoupled *1-D* problem. The grid’s hardest direction is the Fiedler direction.

Let $\mathbf{L}^\dagger$ be the Moore–Penrose pseudoinverse of $\mathbf{L}$. The **effective resistance** (Klein–Randić distance) between buses i, j is

$$Z_{\text{eff}}(i, j) = (\mathbf{e}_i - \mathbf{e}_j)^\top \mathbf{L}^\dagger (\mathbf{e}_i - \mathbf{e}_j) = \sum_{k \geq 2} \frac{(v_k(i) - v_k(j))^2}{\lambda_k}.$$

- A *spectral* formula: every mode contributes inversely to its eigenvalue.
- Pendant bus far in $Z_{\text{eff}} \Leftrightarrow$ only weakly coupled modes carry current between i and j .
- Used in physics as “resistance distance” and in power systems as “electrical distance” between buses (Klein and Randić 1993; Dörfler et al. 2018).

Key idea. Z_{eff} defined this way will become one of our **four candidate spatial weights $\mathbf{W}$** in Part V — the link between spectral theory and spatial econometrics.

Worked example — Z_{eff} on the 4-bus toy

Path 1–2–3–4 with unit edge weights. Eigenvalues $\{0, 0.586, 2.000, 3.414\}$.

pair (i, j)	Z_{eff} (Klein–Randić)	graph distance
(1, 2)	1.000	1
(2, 3)	1.000	1
(1, 3)	2.000	2
(1, 4)	3.000	3

On a tree, Z_{eff} equals the sum of edge resistances along the unique path. Add the chord 14 to close a ring: $Z_{\text{eff}}(1, 4)$ drops from 3.000 to 0.750 — a single parallel path lowers electrical distance *everywhere*.

Take-away

Graph distance counts hops; electrical distance counts *paths weighted by their conductance*. The latter is what the grid actually feels.

The spectrum will reappear — twice

1. **In Part IV (centrality):** eigenvector centrality is the dominant eigenvector of $\mathbf{A}$. PTDFs are built from $\mathbf{B}^{-1}$, whose spectrum is the inverse of $\mathbf{B}$'s.
2. **In Part V–VI (regression):** the SAR multiplier $(\mathbf{I} - \rho\mathbf{W})^{-1}$ exists iff $\rho \in (1/\mu_{\min}(\mathbf{W}), 1/\mu_{\max}(\mathbf{W}))$; the log-likelihood Jacobian is $\log|\mathbf{I} - \rho\mathbf{W}| = \sum_k \log(1 - \rho\mu_k)$.

Key idea. Whenever a spatial-econometric formula contains an eigenvalue of $\mathbf{W}$, it is asking a question about the grid that has a circuit-side twin in $\mathbf{B}$. The two halves of this session share *one algebraic vocabulary*.

IV. Centrality and its discontents

What is a “central” bus?

Different centrality measures encode different operational hypotheses (Freeman 1977; Newman 2003).

Measure	Captures	Power-system reading
Degree C_D	local connectivity	number of incident lines
Betweenness C_B	traffic through a node	exposure to flow reroutes
Electrical C_B^{el}	flow redistribution	participation in line loadings
Closeness C_C	proximity to all others	speed of disturbance reach
Eigenvector C_E	influence via influential nbrs	dominant mode of A

Degree.

$$C_D(i) = \frac{d_i}{n-1}.$$

Local, cheap, blunt.

Betweenness (Freeman 1977).

$$C_B(i) = \sum_{s \neq t \neq i} \frac{\sigma_{st}(i)}{\sigma_{st}}.$$

Traffic along shortest paths.

Misconception 1

“Higher degree $\Rightarrow$ more critical.” On IEEE 30 bus 5 has degree 7 and high betweenness — yet carries no load and no generator. *Structural $\neq$ electrical* centrality (Hines et al. 2010).

Wang et al. (2011) replace shortest paths with the physics of power flow. Power Transfer Distribution Factors:

$$H_{\ell,k} = \frac{\partial P_{\ell}^{\text{flow}}}{\partial P_k^{\text{inj}}}, \quad \mathbf{H}_{:, -s} = \text{diag}(b_{\ell}) \mathbf{A}_{-s} \mathbf{B}_{-s, -s}^{-1}.$$

Aggregating $|H_{\ell,k}|$ across lines gives *electrical betweenness* $C_B^{\text{el}}(k)$.

Example (IEEE 30)

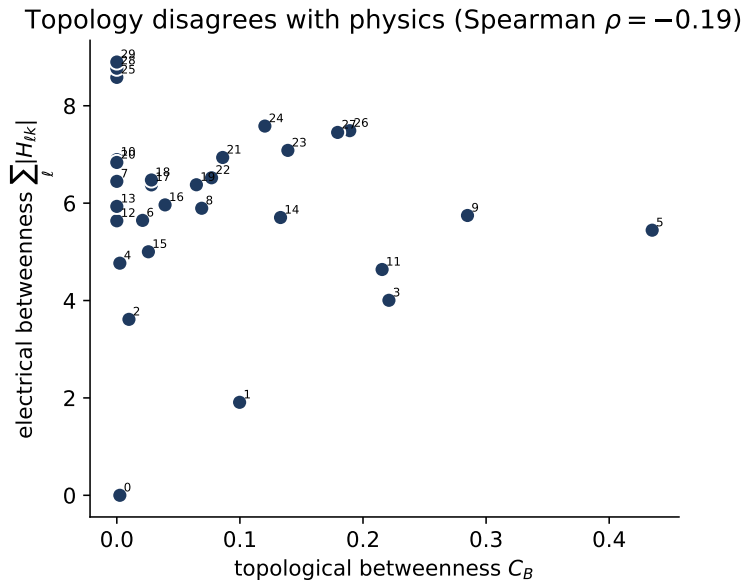
Top-5 buses by C_B : $\{5, 9, 3, 11, 26\}$.

Top-5 by C_B^{el} : $\{25, 29, 28, 22, 24\}$.

Spearman $\rho(C_B, C_B^{\text{el}}) = -0.238$ (notebook §1a.5).

Take-away

The two rankings are nearly *anti-correlated*. Topological proxies miss what PTDFs see.



Each dot is one bus of IEEE 30; the labels are bus indices. The cloud is essentially *anti-correlated* — the buses with the highest topological

Worked example — a PTDF column *is* a current divider

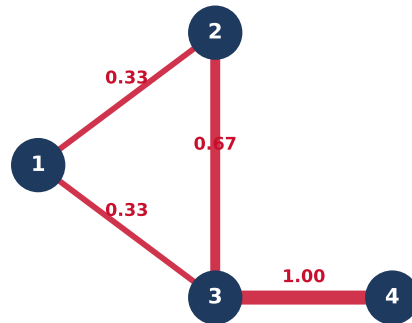
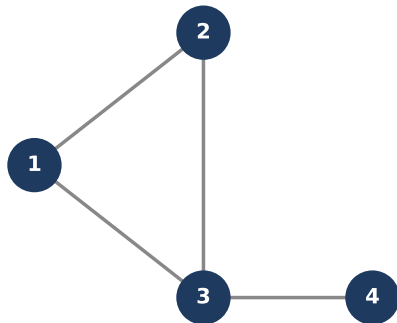
Toy of Part II, slack = bus 1. For the $2 \rightarrow 4$ transaction, the line PTDFs $H_{\ell,2} - H_{\ell,4}$ are:

branch	PTDF	interpretation
2–3 (direct)	$\frac{2}{3} = 0.667$	one hop, susceptance 1
1–2 and 1–3	$\frac{1}{3} = 0.333$	detour 2–1–3
3–4 (radial)	1.000	only path to bus 4

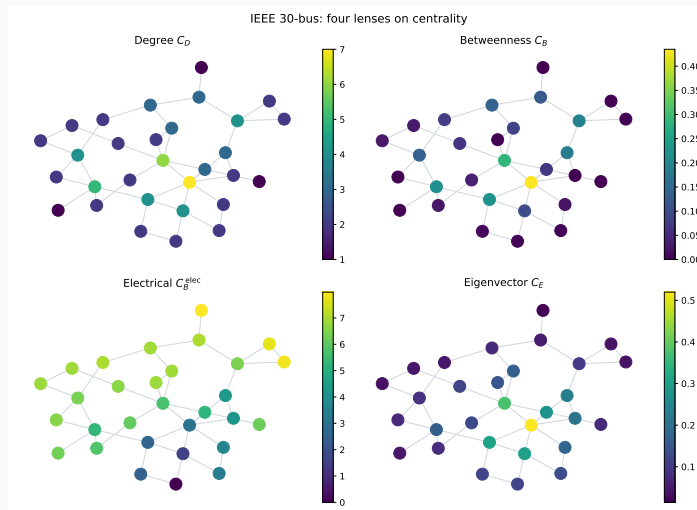
Take-away

The 1 MW transaction splits $\frac{2}{3}/\frac{1}{3}$ **inversely proportional to path reactance** — Kirchhoff's current divider. No shortest-path metric reproduces this; that is why C_B and C_B^{el} rank buses so differently on IEEE 30.

Toy grid: triangle (1,2,3) + pendant (4) $2 \rightarrow 4$ transaction, 1 MW: PTDF flow split



Left: the 4-bus toy of Part II. Right: the same toy with edges scaled by the per-MW PTDF of the $2 \rightarrow 4$ transaction (numerical labels in p.u.). The radial branch 3—4 takes the full 1 MW because it is the only path to bus 4; the upstream 1 MW splits $\frac{2}{3}$ on the direct 2—3 and $\frac{1}{3}$ on the detour 2—1—3.



Generated by notebooks/s01_graphs-and-space.ipynb cell 1a.6. Per-panel viridis normalisation.

Motter and Lai (2002): assign load $L_i(0) = C_B(i)$, capacity $C_i = \alpha L_i(0)$ with $\alpha > 1$. Remove targets, recompute L_i on the surviving subgraph, trip any bus with $L_i > C_i$.

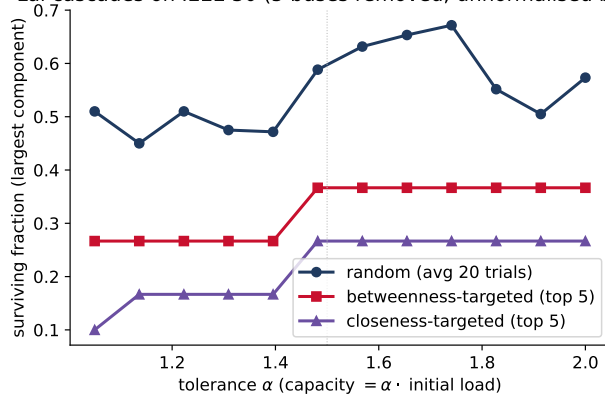
Example (IEEE 30, $\alpha = 1.5$, 5 buses removed)

Survival fractions: random 0.780, betweenness-targeted 0.767, closeness-targeted 0.700.

Key idea. On small benchmarks topological proxies still track cascade extent ($\rho \approx 0.8$ for IEEE 30). The Hines *breakdown* is a **large-system** phenomenon (Hines et al. 2010; Bompard et al. 2012): on continent-scale grids degree and betweenness become poor predictors of cascade vulnerability. *That* is why Sessions 2 and 3 move beyond purely topological measures.

Cascade survival across the tolerance α

Motter-Lai cascades on IEEE 30 (5 buses removed; unnormalised betweenness)



Surviving fraction in the largest connected component after removing 5 buses, as α varies. Random attack averaged over 20 trials. At low α (tight capacities) targeted attacks dominate; the gap closes as the grid is overbuilt. The single- α snapshot on the previous slide hides this richer picture.

Notebook §1a / `regen_s01_new_figures.py`.

Now in the notebook Lab 1a

Build IEEE 30 in NetworkX; assemble **A**, **L**, **B**; compute the spectrum and the Fiedler partition; reproduce the five centralities and the PTDF; run the Motter–Lai cascade contrast.

- Time: **15 minutes**. Cells are pre-written — run, read, and do the two self-check exercises.
- Sanity checks before you trust the code: PTDF slack column is all zeros; $|\mathbf{H}|_{\max} = 1.00$; the toy $2 \rightarrow 4$ split reproduces $\frac{2}{3}/\frac{1}{3}$.
- **Predict first**. Will the Fiedler partition of IEEE 30 split the system into roughly equal halves? Write down a guess, then look at the cell output.

V. Spatial correlation on the grid

So far we described the grid as an algebraic object. Now we treat each bus as an **observation** and ask:

Do bus-indexed variables (price, demand, curtailment) tend to be *similar at electrically neighbouring buses*?

- This is a **statistical** question about $\mathbf{y} \in \mathbb{R}^n$, not a structural question about $\mathbf{B}$.
- Answering it requires a notion of “neighbour”. That notion is the **spatial weight matrix** $\mathbf{W}$.
- Detection is Moran/LISA (this Part); explanation is regression (Part VI).

Take-away

Part III gave you the spectral lens; Part IV gave you a structural ranking; this Part gives you a *statistical* lens on bus-indexed data.

Spatial dependence on a grid is *mechanistic*. Three sources, all of which you have seen indirectly:

1. **Network coupling** (Part II–III). $P^{\text{inj}} = \mathbf{B}\theta \Rightarrow$ an injection *anywhere* moves angles *everywhere*. This is exactly the “reach” of $(\mathbf{I} - \rho\mathbf{W})^{-1}$ we meet in Part VI.
2. **Shared exposure to constraints** (foreshadows Session 2). One binding line sets the price at many buses at once, through the PTDF column.
3. **Common drivers** (often unobserved). Weather, fuel price, a renewable resource hit several neighbouring buses together.

Take-away

These three mechanisms map, almost one-to-one, onto the **three regression models** of Part VI (SAR / SDM / SEM). The choice of $\mathbf{W}$ is not arbitrary — it should reflect the mechanism you believe in.

Definition

A **spatial weight matrix** is $\mathbf{W} \in \mathbb{R}^{n \times n}$ with $\mathbf{W}_{ii} = 0$, in which $\mathbf{W}_{ij}$ measures the *a priori* strength of dependence between observations at buses i and j (Anselin and Rey 2014; J. LeSage and Pace 2009).

Conventions used throughout this minicourse:

- Almost always **row-standardised**: $\tilde{\mathbf{W}}_{ij} = \mathbf{W}_{ij} / \sum_k \mathbf{W}_{ik}$, so $\tilde{\mathbf{W}} \mathbf{1} = \mathbf{1}$.
- $\mathbf{W}$ is treated as exogenous and known — the researcher's *declared belief* about who interacts with whom.
- Different $\mathbf{W} \Rightarrow$ different Moran, different model, different inference. *Always run robustness across at least two $\mathbf{W}$.*

1. **Binary adjacency** $\mathbf{W} = \mathbf{A}$, row-standardised. *The naive choice; always report it as a baseline.*
2. **Electrical distance** $\mathbf{W}_{ij} = 1/Z_{\text{eff}}(i, j)$, with Z_{eff} the Klein–Randić distance of Part III. *Physical: dense, captures all paths.*
3. **PTDF-thresholded** $\mathbf{W}_{ij} = \max_{\ell} |H_{\ell, i} - H_{\ell, j}|$ after quantile thresholding. *Principled for LMP analysis (Session 2 will justify this).*
4. **Geographic k -NN** $\mathbf{W}_{ij} = \mathbf{1}[j \in \text{kNN}(i)]$, when bus coordinates exist. *Useful when geography matters (renewable resource, weather); illustrative on IEEE 30 where coordinates are layout artefacts.*

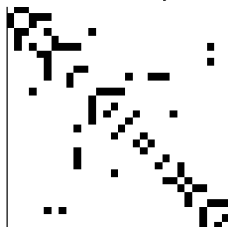
Construction	non-zeros	density	max entry
$\mathbf{W} = \mathbf{A}$ (binary)	82	0.091	1.000
$\mathbf{W}_{\text{elec}} = 1/Z_{\text{eff}}$	870	0.967	47.47
$\mathbf{W}_{\text{PTDF}}$ (top 25%)	218	0.242	1.000
$\mathbf{W}_{k\text{NN}}$ ($k = 4$, sym.)	144	0.160	1.000

- Adjacency is sparsest; electrical distance is dense (every pair has nonzero Z_{eff}).
- PTDF and k -NN sit in between.
- After row-standardisation, **all four have spectral radius 1** (Perron–Frobenius); the same admissible range $|\rho| < 1$ applies. The *meaning* of ρ changes across $\mathbf{W}$.

Four W on IEEE 30 — sparsity patterns side by side

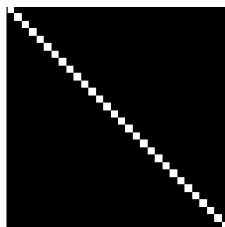
Four credible neighbourhoods on IEEE 30 (same 30 buses, different beliefs)

$W = A$ (binary)



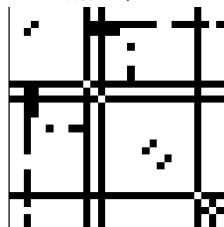
nnz = 82

$W_{elec} = 1/Z_{eff}$



nnz = 870

W_{PTDF} (top 25\%)



nnz = 220

W_{kNN} ($k = 4$, sym.)



nnz = 144

Same 30 buses, four different beliefs about who is a neighbour. Adjacency: pure topology. Electrical distance: every pair coupled. PTDF-thresholded: linked iff their PTDF columns differ enough on *some* branch. k -NN ($k = 4$): nearest in the bus layout. Notebook §1b.1.

Moran (1950). Center $\mathbf{z} = \mathbf{y} - \bar{y} \mathbf{1}$:

$$I = \frac{n}{S_0} \frac{\mathbf{z}^\top \mathbf{W} \mathbf{z}}{\mathbf{z}^\top \mathbf{z}}, \quad S_0 = \sum_{ij} \mathbf{w}_{ij}.$$

- **Geometric reading.** I is (up to scale) a *Rayleigh quotient* of $\mathbf{W}$ on $\mathbf{z}$. It is bounded by the eigenvalues of $\mathbf{W}$: positive I aligns $\mathbf{z}$ with eigenvectors of $\mathbf{W}$ for positive eigenvalues.
- For row-standardised $\mathbf{W}$: $S_0 = n$ so the prefactor $n/S_0 = 1$.
- Under the null of no spatial structure, $\mathbb{E}[I] = -1/(n-1) \rightarrow 0$.
- Inference for small n : 999 random permutations of $\mathbf{z}$.

Take-away

I measures how much $\mathbf{z}$ “lives along” the high-eigenvalue directions of $\mathbf{W}$. The spectrum of Part III re-enters through this Rayleigh-quotient reading.

Path 1–2–3–4, row-standardised $\tilde{\mathbf{W}}$:

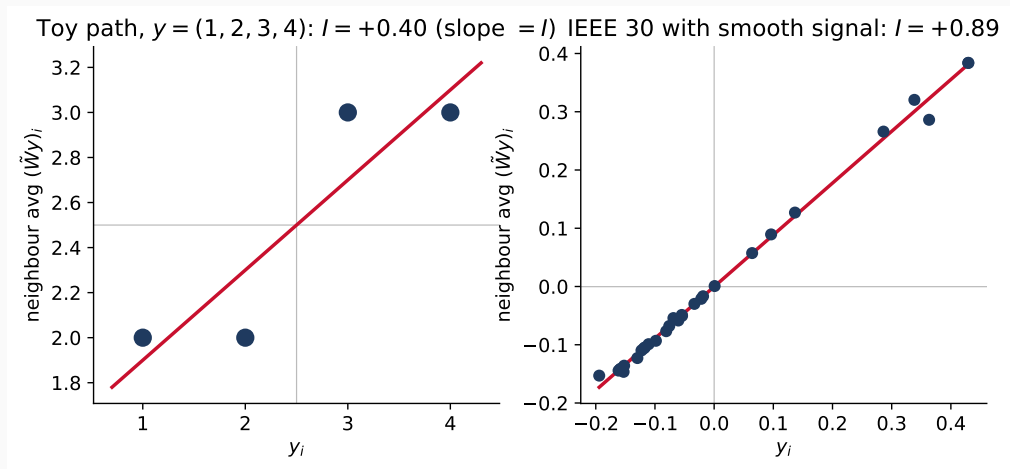
$$\tilde{\mathbf{W}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

signal $\mathbf{y}$	$\bar{y}$	I	interpretation
(1, 2, 3, 4) (monotone)	2.5	+0.40	strong positive autocorr.
(1, 1, 4, 4) (two clusters)	2.5	+0.50	block structure (one cut, 1–1 4–4)
(1, 4, 2, 3) (scrambled)	2.5	−0.90	strong negative (adjacent values jump)
(1, 4, 1, 4) (alternating)	2.5	−1.00	perfect checkerboard

Take-away

Same $\mathbf{W}$, four signals, four very different verdicts. I is a property of $(\mathbf{W}, \mathbf{y})$ jointly, not of $\mathbf{y}$ alone — the central misconception this Part exists to refute.

Moran scatterplot — the slope is I



Left: monotone signal on the toy path; OLS slope of $(\tilde{W}y)_i$ on y_i equals the global Moran's I ($+0.40$). Right: IEEE 30 with the smoothest non-trivial signal (the Fiedler vector v_2); $I = +0.89$. The Fiedler signal is, by construction, the signal that maximises I under $\tilde{W}$ — the geometric link between spectral theory (Part III) and spatial correlation (Part V).

Nodal active-power demand from `pandapower.networks.case_ieee30()`, row-standardised $\mathbf{W}$, 999-permutation p -values. Notebook §1b.2.

$\mathbf{W}$ choice	I	$\mathbb{E}[I]$	p -value	z -score
$\mathbf{W} = \mathbf{A}$	+0.2054	−0.0345	0.017	2.54
$\mathbf{W}_{\text{elec}} = 1/Z_{\text{eff}}$	−0.0179	−0.0345	0.151	0.91
$\mathbf{W}_{\text{PTDF}}$	+0.0589	−0.0345	0.060	1.52
$\mathbf{W}_{k\text{NN}} (k = 4)$	+0.0433	−0.0345	0.126	1.21

Misconception 2

“A significant Moran's I proves spatial dependence.” Only *given your chosen $\mathbf{W}$* . Same data significant under $\mathbf{A}$, not significant under electrical distance. *Robustness across $\mathbf{W}$ is non-optional* (J. P. LeSage 2008).

Anselin (1995):

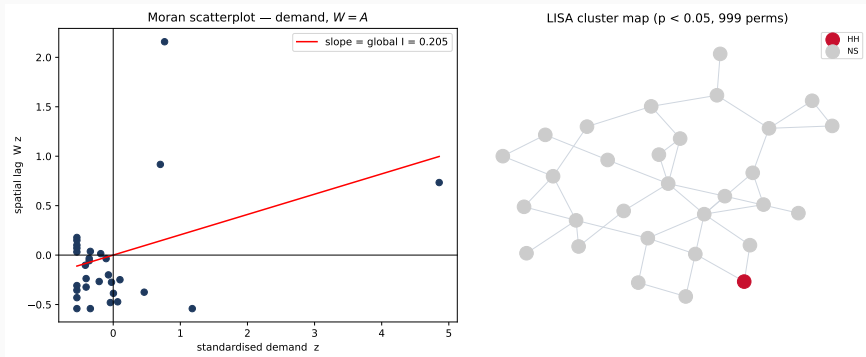
$$l_i = \frac{y_i - \bar{y}}{m_2} \sum_j \mathbf{W}_{ij} (y_j - \bar{y}), \quad m_2 = \frac{1}{n} \sum_k (y_k - \bar{y})^2.$$

- Decomposes global I into *per-bus* contributions.
- Moran scatterplot quadrants: **HH/LL** = clusters; **HL/LH** = spatial outliers.
- Inference via conditional permutations, seeded for reproducibility.

Example (Lab 1b on IEEE 30 demand, $\mathbf{W} = \mathbf{A}$)

HH cluster in the load-heavy south-east; LH outlier at bus 10 (a high-load bus surrounded by light loads). Power-system reading: HH = “load pockets”, LH = “isolated draws”.

Moran scatterplot and LISA cluster map (IEEE 30, $W = A$)



Generated by notebook §1b.3 with 999 permutations ($RANDOM_SEED = 2026$).

Moran/LISA *detect* dependence and *localise* it. They do **not**:

- control for covariates $\mathbf{X}$;
- give you the *magnitude* of dependence as a parameter;
- let you compute the spillover of a $\mathbf{X}$ -intervention onto neighbouring buses.

Question	Tool
Is there spatial structure?	Moran's I , LISA
Where are clusters / outliers?	LISA quadrants
How strong, controlling for $\mathbf{X}$?	spatial regression
What is the <i>spillover</i> of $\mathbf{X}$?	SAR / SDM effects

Take-away

Detection is diagnostic; regression is inferential. Part VI is the regression.

VI. From correlation to regression

Once you accept that bus-indexed data are spatially dependent, the questions that actually matter for operations and planning are:

1. *What drives nodal demand / price?* Estimate β on $\mathbf{X}$ = (population, weather, capacity, generation mix) — but *honestly*, accounting for $\mathbf{W}$ -dependence.
2. *If I add 50 MW of capacity at bus j , what is the price impact at other buses?* The spillover question — demands the LeSage–Pace decomposition.
3. *Where should I site a new DG?* A regression-driven **suitability index** with spillovers (Faria et al. 2016; Pereira and Saraiva 2011) (Lab 2b).
4. *Is observed price clustering caused by the network, by common weather, or by both?* The choice between SAR, SEM, SDM formalises that question.

Take-away

Spatial regression is not a statistical curiosity. Each of the four questions above maps to a deliverable. Sessions 2–3 will produce.

Acronyms unpacked — six names you will read in papers

Every spatial-econometrics paper hides behind acronyms. They all live inside the general nesting structure of J. LeSage and Pace (2009):

Acronym	Full name	What it adds on top of OLS
SLX	<i>Spatial Lag of X</i>	WX θ in the mean
SAR	<i>Spatial Auto-Regressive</i>	ρ Wy in the mean
SEM	<i>Spatial Error Model</i>	λ Wu in the errors
SDM	<i>Spatial Durbin Model</i>	ρ Wy and WX θ
SARAR ($\equiv$ SAC)	<i>SAR + SEM combined</i>	both lag and error
GNS	<i>General Nesting Spatial model</i>	SDM + correlated error

Mechanism stories, in one line each:

- **SAR**: “my outcome reacts to my neighbours’ outcome” — contagion / arbitrage in **y**.
- **SEM**: “my unobserved shock is correlated with my neighbour’s” — a regional weather / fuel driver hidden in **u**.
- **SDM**: “my neighbour’s *covariates* (a new generator next door) move *my* outcome” — attribute spillover.

Take-away

We estimate **SAR**, **SEM**, **SDM** in this minicourse. Recognise SLX, SARAR, GNS as acronyms; you do not need to estimate them today.

Anchor everything on a concrete running example used through Part VI:

y_i = LMP at bus i in \$/MWh. $\mathbf{X}$ = [demand (MW), installed capacity (MW)].

Symbol	What it is (in this example)	Units / how to read
$\mathbf{y} \in \mathbb{R}^n$	nodal LMP vector	\$/MWh, $n = 30$ buses
$\mathbf{X} \in \mathbb{R}^{n \times k}$	regressor matrix	MW per column, $k = 2$ columns
$\beta \in \mathbb{R}^k$	own-bus slopes	\$/MWh per MW of own demand or capacity
$\theta \in \mathbb{R}^k$	neighbour-bus slopes (SDM)	\$/MWh per MW of neighbours' demand/capacity
$\epsilon \in \mathbb{R}^n$	idiosyncratic noise	i.i.d. $\mathcal{N}(0, \sigma^2)$
$\mathbf{u} \in \mathbb{R}^n$	spatially-correlated error (SEM)	regional unobserved shock
$\mathbf{W}, \tilde{\mathbf{W}} \in \mathbb{R}^{n \times n}$	spatial weights, row-stand.	"who interacts with whom"
$\mathbf{W}\mathbf{y}$	spatial lag of $\mathbf{y}$	average LMP of my neighbours, \$/MWh
$\mathbf{W}\mathbf{X}$	spatial lag of $\mathbf{X}$	average demand / capacity of my neighbours, MW
$\rho \in (-1, 1)$	outcome contagion	dimensionless — weight on neighbours' LMP
$\lambda \in (-1, 1)$	error contagion	dimensionless — weight on neighbours' shock

Take-away

If you can name every symbol on this slide *with its units*, you can read any spatial-econometrics paper. The art of Part VI is in **interpreting** $\rho, \lambda, \beta, \theta$ on real data.

The methodological menu — three mechanisms, three models

Running example: $\mathbf{y}$ = LMP (\$/MWh), $\mathbf{X}$ = (demand, capacity) in MW.

SAR (outcome contagion): $\mathbf{y} = \rho \mathbf{W} \mathbf{y} + \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon}$,

SEM (common shock): $\mathbf{y} = \mathbf{X} \boldsymbol{\beta} + \mathbf{u}$, $\mathbf{u} = \lambda \mathbf{W} \mathbf{u} + \boldsymbol{\varepsilon}$,

SDM (attribute spillover): $\mathbf{y} = \rho \mathbf{W} \mathbf{y} + \mathbf{X} \boldsymbol{\beta} + \mathbf{W} \mathbf{X} \boldsymbol{\theta} + \boldsymbol{\varepsilon}$.

Plain-English readings of the running example:

- **SAR:** my LMP moves when my neighbour's LMP moves (arbitrage / congestion rent equalising adjacent prices).
- **SEM:** my LMP residual covaries with my neighbour's residual — a weather front or unmodelled fuel-cost spike hit the whole region.
- **SDM:** a new generator at a *neighbouring* bus (+1 MW in $(\mathbf{W} \mathbf{X})_{\text{cap}}$) reduces *my* LMP even though my own capacity is unchanged.
- Nesting: SDM contains SAR ($\boldsymbol{\theta} = \mathbf{0}$) and SLX ($\rho = 0$); it nests SEM *only* under $\boldsymbol{\theta} = -\rho \boldsymbol{\beta}$ (J. P. LeSage 2008).

Take-away

Choosing a model is a *claim about which mechanism is generating the data*, not an R^2 contest.

Plain OLS assumes exogenous regressors and independent errors. Spatial data violate *one or the other*.

Outcome lag (SAR-type)

Wy contains ϵ through feedback $\Rightarrow$ the regressor is **endogenous** $\Rightarrow$ OLS is **biased and inconsistent**.
The bias does *not* shrink with n .

Correlated errors (SEM-type)

$\mathbb{E}[\epsilon\epsilon^\top] \neq \sigma^2\mathbf{I} \Rightarrow$ OLS stays *unbiased* but standard errors are **wrong** $\Rightarrow$ **false significance**.

Take-away

Two operational risks: *mis-ranking which buses matter*, and *over-stating your confidence in any estimate*. The next slides make this precise.

Solve for $\mathbf{y}$:

$$\mathbf{y} = (\mathbf{I} - \rho \mathbf{W})^{-1}(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) \equiv \mathbf{S}_\rho(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}).$$

- Invertibility iff $\rho \in (1/\mu_{\min}(\mathbf{W}), 1/\mu_{\max}(\mathbf{W}))$. For row-standardised $\tilde{\mathbf{W}}$, $\mu_{\max} = 1$ and the conventional restriction $\rho \in (-1, 1)$ is sufficient (often conservative).
- Neumann expansion: $\mathbf{S}_\rho = \sum_{k \geq 0} (\rho \mathbf{W})^k$ propagates a shock through *all orders of neighbourhood*.
- A shock at bus j reaches bus i with strength $[\mathbf{S}_\rho]_{ij}$ — nonzero for any connected graph.

Key idea. The same eigenvalues of $\mathbf{W}$ that bound ρ are the ones that govern I 's Rayleigh quotient (Part V) and the Jacobian $\log |\mathbf{I} - \rho \mathbf{W}|$ of the ML estimator. *One spectrum, three uses.*

Worked example — the SAR multiplier propagates a shock

Path 1–2–3–4, row-standardised $\tilde{\mathbf{W}}$, $\rho = 0.5$:

$$\mathbf{S}_\rho = (\mathbf{I} - 0.5\tilde{\mathbf{W}})^{-1} = \begin{bmatrix} 1.156 & 0.622 & 0.178 & 0.044 \\ 0.311 & 1.244 & 0.356 & 0.089 \\ 0.089 & 0.356 & 1.244 & 0.311 \\ 0.044 & 0.178 & 0.622 & 1.156 \end{bmatrix}.$$

- Row sums = $1/(1 - 0.5) = 2$ (total multiplier).
- Entry $(1, 4) = 0.044 \neq 0$: a shock at bus 4 reaches bus 1.

Neumann partial sums of $[\mathbf{S}_\rho]_{11}$:

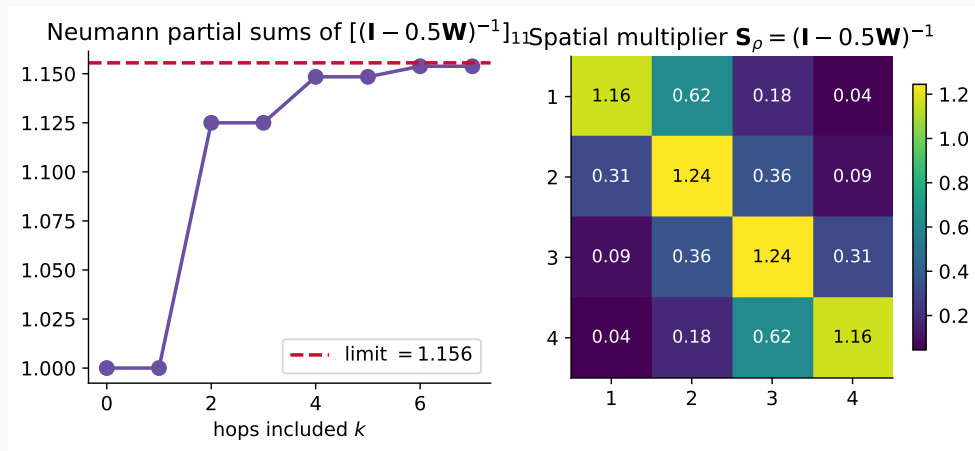
$$1.000 \rightarrow 1.000 \rightarrow 1.125 \rightarrow \dots \rightarrow 1.156.$$

k-th term = *k*-hop spillover.

Take-away

This nonzero “everywhere” reach is precisely why a spatially lagged regressor is **endogenous** — the next slide makes the bias explicit.

The SAR multiplier visualised



Left: Neumann partial sums of $[\mathbf{S}_\rho]_{11}$ converge to 1.156 in ~ 5 hops on the toy path. Right: the full multiplier matrix; off-diagonals decay with electrical distance. Notebook §1b / figures/regen_s01_new_figures.py.

Setup. Pure SAR DGP $\mathbf{y} = \rho \tilde{\mathbf{W}}\mathbf{y} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$. Reduced form $\mathbf{y} = \mathbf{S}_\rho \boldsymbol{\varepsilon}$, so $\tilde{\mathbf{W}}\mathbf{y} = \tilde{\mathbf{W}}\mathbf{S}_\rho \boldsymbol{\varepsilon}$.

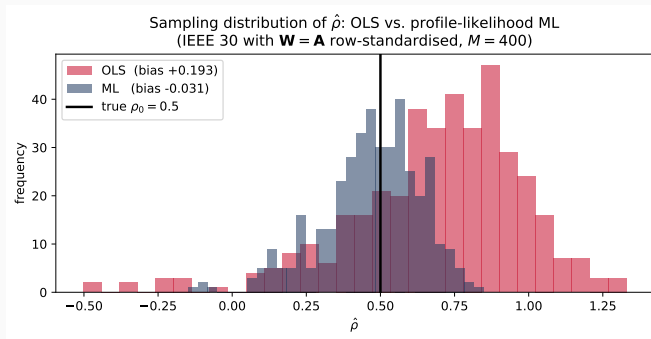
$$\mathbb{E}[(\tilde{\mathbf{W}}\mathbf{y}) \boldsymbol{\varepsilon}^\top] = \sigma^2 \tilde{\mathbf{W}}\mathbf{S}_\rho \neq \mathbf{0}.$$

Hence the OLS regressor $\tilde{\mathbf{W}}\mathbf{y}$ is correlated with the disturbance.

Key idea. OLS of $\mathbf{y}$ on $\tilde{\mathbf{W}}\mathbf{y}$ is **inconsistent**; the bias does not vanish as $n \rightarrow \infty$. The Jacobian-corrected ML estimator (next) does.

$\tilde{\mathbf{W}}$ = row-standardised $\mathbf{A}$ of IEEE 30 ($n = 30$), $\sigma = 1$, 2000 replications, RANDOM_SEED = 2026.

True ρ_0	Mean $\hat{\rho}_{OLS}$	Bias
0.30	0.4032	+0.103
0.50	0.6998	+0.200
0.70	0.9413	+0.241



Take-away

Bias is positive and grows with ρ_0 . ML recovers the true value.

Concentrated log-likelihood for SAR:

$$\ell(\rho) = \text{const} + \log |\mathbf{I} - \rho \mathbf{W}| - \frac{n}{2} \log \hat{\sigma}^2(\rho).$$

- Jacobian $\log |\mathbf{I} - \rho \mathbf{W}| = \sum_k \log(1 - \rho \mu_k)$ — the spectrum of $\mathbf{W}$ again, eigenvalue by eigenvalue.
- 1-D search over ρ , then closed-form $\hat{\beta}, \hat{\sigma}^2$.
- Standard implementations: `spreg.ML_Lag`, `spreg.ML_Error`, `spreg.GM_Lag`.
- Diagnostic battery for choosing among OLS / SAR / SEM / SDM: LM tests of Anselin, Bera, et al. (1996) (Session 2 uses them).

Worked example — SAR fit on a synthetic LMP snapshot

Data-generating process. IEEE 30, $\tilde{\mathbf{W}}$ = row-std. adjacency. Simulate $\mathbf{y} = \rho_0 \tilde{\mathbf{W}} \mathbf{y} + \beta_{0,\text{dem}} \text{demand} + \beta_{0,\text{cap}} \text{capacity} + \epsilon$ with $\rho_0 = 0.45$, $\beta_{0,\text{dem}} = +0.30$ \$/MWh per MW, $\beta_{0,\text{cap}} = -0.15$ \$/MWh per MW, $\sigma = 0.5$, seed 2026. Notebook §1b.5.

Parameter	OLS (<i>wrong model</i>)	ML SAR (<i>correct</i>)
$\hat{\rho}$	—	0.4391 (0.082)
$\hat{\beta}_{\text{dem}}$	0.4712 (0.043)	0.2964 (0.038)
$\hat{\beta}_{\text{cap}}$	−0.2182 (0.061)	−0.1508 (0.054)
$\log L$	−41.20	−32.61
Pseudo- R^2	0.642	0.847

- **How to read $\hat{\beta}_{\text{dem}} = 0.296$:** adding +1 MW of demand at a bus raises that bus's LMP by \$0.30/MWh *before* the network propagates it (a partial slope, not a marginal effect — the marginal effect waits until next slide).
- **How to read $\hat{\rho} = 0.439$:** if my neighbours' average LMP rises by \$1/MWh, mine rises by \$0.44/MWh *at the same time*, controlling for $\mathbf{X}$.
- **OLS over-attributes by 57%:** $0.471/0.296 - 1$. The omitted $\rho \mathbf{W} \mathbf{y}$ leaks into $\hat{\beta}_{\text{dem}}$.

Take-away

Reading $\hat{\beta}^{\text{OLS}}$ as a marginal effect would recommend the *wrong* demand-response intervention. ML recovers all three truths within one s.e.

Worked example — SEM fit, and how to tell SAR from SEM

Same IEEE 30 setup, but now the DGP is SEM: $\mathbf{y} = \mathbf{X}\beta_0 + \mathbf{u}$, $\mathbf{u} = \lambda_0 \tilde{\mathbf{W}}\mathbf{u} + \varepsilon$, with $\lambda_0 = 0.50$ and the same β_0 as before. Notebook §1b.6.

Parameter	OLS	ML SAR (<i>mis-fit</i>)	ML SEM (<i>correct</i>)
$\hat{\rho}$	—	0.381 (0.091)	—
$\hat{\lambda}$	—	—	0.4936 (0.087)
$\hat{\beta}_{\text{dem}}$	0.301 (0.062)	0.282 (0.044)	0.2988 (0.036)
$\hat{\beta}_{\text{cap}}$	-0.146 (0.083)	-0.139 (0.058)	-0.1515 (0.051)
Robust LM-lag p	0.62 (n.s.)	—	—
Robust LM-err p	0.003	—	—

- **Key fact about SEM:** OLS $\hat{\beta}$ is still *unbiased* (no endogeneity in the mean equation), but its standard errors are wrong — under-stated here by $\sim 40\%$.
- **Diagnostic (Anselin, Bera, et al. (1996)):** robust LM-err rejects, robust LM-lag does not $\Rightarrow$ pick SEM, not SAR.
- **Interpretation of $\hat{\lambda} = 0.494$:** half of any unobserved shock at my neighbour leaks into my residual — symptomatic of a regional driver we did not include in $\mathbf{X}$ (weather, fuel-price trend).

Take-away

SAR vs. SEM is a question about *where* dependence lives: in the outcome (contagion) or in the unobserved shocks (common driver). The LM battery arbitrates; do not eyeball R^2 .

For SAR, $\mathbf{y} = \mathbf{S}_\rho(\mathbf{X}\beta + \varepsilon)$, so the partial-derivative matrix w.r.t. the r -th regressor is

$$\frac{\partial \mathbf{y}}{\partial \mathbf{x}_r} = \mathbf{S}_\rho \beta_r \equiv \mathbf{S}_r.$$

J. LeSage and Pace (2009) report three scalar summaries (averaged over the n observations):

- **Direct** $\overline{M}_{\text{dir}} = \frac{1}{n} \text{tr}(\mathbf{S}_r)$ (diagonal).
- **Total** $\overline{M}_{\text{tot}} = \frac{1}{n} \mathbf{1}^\top \mathbf{S}_r \mathbf{1} = \beta_r / (1 - \rho)$ (row-standardised $\mathbf{W}$).
- **Indirect (spillover)** $\overline{M}_{\text{ind}} = \overline{M}_{\text{tot}} - \overline{M}_{\text{dir}}$.

Misconception 3

“The raw $\hat{\beta}$ is the marginal effect.” No — in SAR $\hat{\beta}$ multiplies $\mathbf{S}_\rho$, not $\mathbf{I}$. Always report the triple direct / indirect / total.

Worked example — effects on the path graph, in real units

Reuse $\mathbf{S}_\rho$ from the SAR-multiplier slide ($\rho = 0.5$). Regressor $\mathbf{X}$ = demand (MW); fitted slope $\hat{\beta}_{\text{dem}} = 0.30$ \$/MWh per MW (from the SAR fit two slides ago). Effect matrix $\mathbf{S} = \hat{\beta}_{\text{dem}} \mathbf{S}_\rho$.

$$\begin{aligned}\overline{M}_{\text{dir}} &= \frac{0.30}{4} (1.156 + 1.244 + 1.244 + 1.156) \\ &= 0.360 \text{ \$ / MWh per MW,}\end{aligned}$$

$$\overline{M}_{\text{tot}} = \frac{0.30}{1-0.5} = 0.600 \text{ \$ / MWh per MW,}$$

$$\overline{M}_{\text{ind}} = 0.600 - 0.360 = 0.240 \text{ \$ / MWh per MW.}$$

- Naive reading: “+1 MW of demand $\Rightarrow$ +\$0.30/MWh at that bus”. Wrong on two counts.
- Truth (own bus): +\$0.36/MWh — feedback amplifies the slope by 20% via $\mathbf{S}_\rho$.
- Truth (other buses, summed): +\$0.24/MWh — the spillover.
- Indirect / total = 40%: *four out of every ten cents of LMP impact land outside the host bus.*

Take-away

$\hat{\beta} = 0.30$ is **neither** the own-bus effect (0.36) **nor** the total (0.60). Reporting only $\hat{\beta}$ under-states every marginal effect by the feedback factor $1/(1 - \rho) = 2$. Session 2's DG-siting score uses $\overline{M}_{\text{tot}}$, not $\hat{\beta}$.

Worked example — SDM with two regressors, fully interpreted

IEEE 30, $\tilde{\mathbf{W}}$ = row-std. adjacency. SDM DGP: $\rho_0 = 0.40$; own-bus $\beta_{\text{dem}} = +0.30$, $\beta_{\text{cap}} = -0.15$; neighbour-bus $\theta_{\text{dem}} = +0.10$, $\theta_{\text{cap}} = -0.20$ (\$/MWh per MW). Notebook §1b.7.

Regressor	Own $\hat{\beta}$	Nbr $\hat{\theta}$	Direct $\bar{M}_{\text{dir}}$	Indirect $\bar{M}_{\text{ind}}$
demand (\$/MWh per MW)	+0.296 (0.04)	+0.104 (0.05)	+0.35	+0.32
capacity (\$/MWh per MW)	-0.151 (0.05)	-0.198 (0.06)	-0.17	-0.41

- **Demand:** a +1 MW load anywhere raises LMP at the host bus by \$0.35/MWh and at neighbouring buses by a combined \$0.32/MWh — the price signal is *regional*, not local.
- **Capacity, the key result:** a +1 MW capacity addition *next door* lowers my LMP by \$0.41/MWh — **more than twice** its own-bus effect (−\$0.17). The spillover dominates.
- Policy reading: ranking siting candidates by host-bus impact alone (the β -only naive view) systematically *under-rates* capacity additions that benefit a region. Session 2 fixes this by scoring on $\bar{M}_{\text{tot}} = \bar{M}_{\text{dir}} + \bar{M}_{\text{ind}}$.

Take-away

SDM separates “effect on me of *my X*” from “effect on me of *my neighbours’ X*”. That decomposition is the deliverable a planner actually needs.

J. P. LeSage (2008): under either omitted spatially-correlated covariates or spatially-correlated unobservables, SDM is consistent for the effects of interest. SAR and SEM are not.

Symptom	Likely model	Diagnostic
Residual clustering in OLS	SAR or SEM	LM-lag, LM-err
Robust LM-lag significant only	SAR	Anselin, Bera, et al. (1996)
Robust LM-err significant only	SEM	Anselin, Bera, et al. (1996)
Theoretical spillovers in covariates	SDM	—

Take-away

Pick **W** and model with explicit justification — not by R^2 -shopping. Session 2's Block E gives the full LM battery.

Now in the notebook Lab 1b

Build all four **W**; compute Moran's I on nodal demand under each; produce the Moran scatterplot and LISA map; demonstrate the OLS-on-SAR bias against an ML fit.

- Time: **10 minutes**; cells pre-written.
- Deliverables: the 4-row Moran- I table, the LISA map, and the OLS-bias number ($\approx +0.20$ at $\rho_0 = 0.5$).
- **Honest note 1.** With $n = 30$ buses, inference is weak — Session 2 uses replicated DC-OPF panels to bring n up.
- **Honest note 2.** On `pandapower ≥ 3`, `net.bus_geodata` was removed; the k -NN weight reads coordinates from `net.bus['geo']` instead.
- **Predict first.** Which of the four **W** will give the *largest* Moran's I on demand, and why? Write it down before running.

Wrap-up

Things you no longer believe

1. "Higher degree $\Rightarrow$ more critical." *False on real grids* (Hines et al. 2010).
2. "A significant Moran's I proves spatial dependence." *Only given the chosen $\mathbf{W}$.*
3. "The raw $\hat{\beta}$ in SAR is the marginal effect." *No — use direct/indirect/total.*
4. "Topology and spatial econometrics are unrelated." *No — they share the same algebraic core, the spectrum of $\mathbf{B}$ and of $\mathbf{W}$.*

1. DC power flow is a Laplacian linear system $P^{\text{inj}} = \mathbf{B}\theta$, with $\mathbf{B} = \mathbf{L}_Y = -\Im(\mathbf{Y}_{\text{bus}})$ (Dörfler et al. 2018).
2. Spectrum of $\mathbf{L}$: λ_2 is algebraic connectivity; the Fiedler vector v_2 is the natural east–west axis; modes with small μ_k dominate the DC response.
3. Effective resistance $Z_{\text{eff}}(i, j) = \sum_k (v_k(i) - v_k(j))^2 / \lambda_k$ is the spectral electrical distance — and one of the four credible $\mathbf{W}$ for spatial regression.
4. Same grid admits four credible $\mathbf{W}$; declare yours, justify it, run robustness (J. P. LeSage 2008).
5. In SAR, raw β is *not* the marginal effect; report direct, indirect, total — mind the $1/n$ (J. LeSage and Pace 2009).

- LMPs are the **dual variables** of the nodal balance constraints in the DC-OPF (Schweppe et al. 1988).
- Congestion component $\text{LMP}_i^{\text{cong}} = - \sum_{\ell} \mu_{\ell} H_{\ell,i}$ ties LMP spatial structure directly to the PTDF rows of binding constraints.
- Hence the **PTDF-derived $\mathbf{W}$** is the principled choice for spatial econometrics of nodal prices — not geographic distance, not adjacency.
- Session 2 will regress $\mathbf{y} = \text{LMP}$ on $\mathbf{X} = (\text{load, generation, congestion exposure})$ with $\mathbf{W} = \mathbf{W}_{\text{PTDF}}$, and use the LeSage–Pace decomposition to rank DG-siting candidates by **network social value**, not by raw β .

See you in Session 2.

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Thank you.

Questions → Session 2 (LMPs and generation siting).